

Performance Analysis of TCP and SaTCP in LEO Satellite Networks: An Emulation Study

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Cover Note: This research paper presents my independent research project conducted in June 2024, where I implemented and analyzed the performance of different transport protocols in Low Earth Orbit (LEO) satellite networks using the LeoEM emulation platform. While the LeoEM platform was originally developed by Xuyang Cao and contributors, I modified its configuration and workflow to suit my specific experimental setup. I was solely responsible for setting up the modified environment, conducting experiments, analyzing the results, and documenting the findings. This work demonstrates my ability to adapt open-source tools for complex systems research and analyze technical results.

Abstract—This paper presents a comparative analysis of the performance of TCP and SaTCP in low Earth orbit (LEO) satellite networks through emulation-based experimentation. Using the open-source LeoEM platform and modified for this study, I evaluated the protocols' behavior under challenging conditions typical of LEO networks, including frequent handover events and variable round-trip times (RTT). My results demonstrate that while SaTCP achieves better RTT stability compared to traditional TCP, it shows unexpected behavior in throughput maintenance, particularly during handover events. The study provides insights into the practical implementation challenges of specialized transport protocols in LEO satellite networks and suggests directions for future improvements in both protocol design and emulation methodologies.

I. INTRODUCTION

LEO satellite networks are becoming increasingly important for global internet connectivity, with companies like Starlink deploying thousands of satellites in low Earth orbit. However, these networks present unique challenges for transport protocols, including frequent handover events, high variability in RTT, and throughput fluctuations. Traditional TCP, designed primarily for terrestrial networks, may not perform optimally in these conditions. SaTCP, a specialized protocol designed for LEO networks, attempts to address these challenges through link-layer integration and predictive mechanisms [2].

II. BACKGROUND AND RELATED WORK

A. Challenges in LEO Communication

LEO satellite networks face several distinct challenges that impact transport protocol performance:

- 1) **Frequent Handover Events:** As satellites move rapidly relative to ground stations, connections must frequently switch between satellites, potentially disrupting data transmission.
- 2) **RTT Variability:** The dynamic nature of satellite positions leads to varying propagation delays, affecting protocol timing mechanisms.
- 3) **Throughput Fluctuations:** Changes in satellite positions and handover events can cause significant variations in available bandwidth.

B. SaTCP Design

SaTCP was developed specifically to address these challenges through two key features [2]:

- 1) **Satellite Location Prediction:** The protocol computes the timing of potential disruptive events and preemptively adjusts its behavior, including freezing the congestion window during handover events.
- 2) **Cross-Layer Integration:** SaTCP incorporates real-time data from the link layer to make informed decisions about congestion control and transmission rates.

III. METHODOLOGY

A. System Architecture

The emulation platform that I used is based on the open source LeoEM [1], originally developed by Xuyang Cao and the contributors. For this study, I customized its configuration to better suit my specific experimental setup. The emulation consists of three main stages, as illustrated in Figure 1:

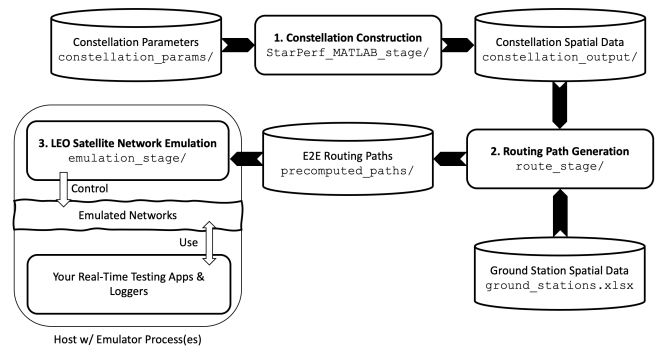


Fig. 1. LeoEM System Architecture [1]

- 1) Constellation Construction: Takes constellation parameters as input and generates spatial data using StarPerf. Data were used from the Starlink constellation that comprises 1,588 satellites in 5,731 cycles.
- 2) Routing Path Generation: Combines constellation data with ground station information to compute inter-satellite link (ISL) paths, specifically focusing on the San Diego to Shanghai route.
- 3) LEO Satellite Network Emulation: Implements the network environment and enables testing applications. I made custom modifications to integrate SaTCP and enhance the logging of handover events.

B. Experimental Setup

The implementation environment consisted of:

- Ubuntu Server 24.04 running in a virtual machine
- Linux kernel version 6.9.6 for SaTCP implementation
- iPerf3 for throughput testing
- TShark for RTT measurement and traffic capture
- Automated handover event logging

IV. RESULTS AND ANALYSIS

A. TCP Performance

1) Round-Trip Time Analysis

Figure 2 shows TCP RTT measurements over time:

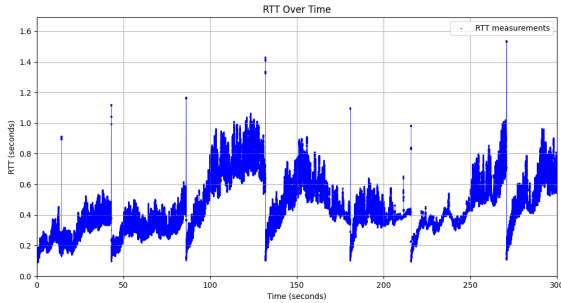


Fig. 2. TCP RTT Over Time

The RTT measurements exhibit significant variation, ranging from 0.2 to 1.6 seconds, with a mean of 0.44 seconds and a standard deviation of 0.31 seconds. This variability poses challenges for real-time applications, which typically require consistent RTT below 300ms for acceptable performance. Figure 3 reveals the correlation between handover events and RTT spikes:

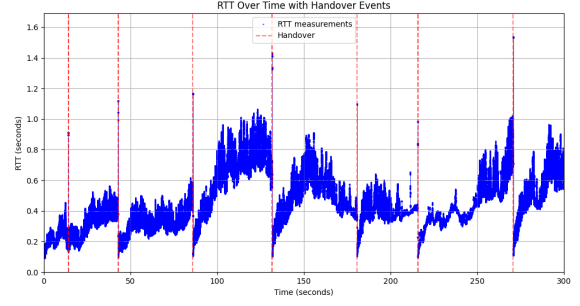


Fig. 3. TCP RTT with Handover Events

During handover events, RTT typically spikes to 1.2-1.4 seconds, representing a 3-4x increase from baseline measurements. These spikes can be attributed to three primary factors:

- Queue buildup during satellite switching as packets are buffered
- Path reconfiguration overhead as new routing tables are computed
- TCP's congestion control mechanism interpreting the transition delay as network congestion

The frequency and magnitude of these RTT spikes suggest that in real-world deployments, users would experience periodic quality degradation in interactive applications. For instance, video conferencing applications would likely show visible freezing or stuttering during these handover events, while gaming applications might become temporarily unresponsive.

2) Throughput Analysis

TCP throughput measurements (Figure 4) demonstrate relatively stable performance between handover events:

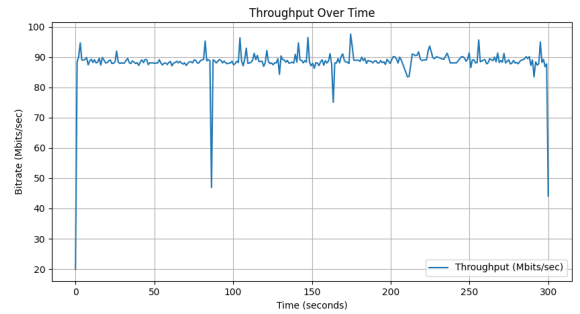


Fig. 4. TCP Throughput Over Time

The baseline throughput averages 89.84 Mbits/sec (std. dev. = 12.46 Mbits/sec), which represents approximately 89.8% utilization of the theoretical 100 Mbits/sec link capacity. This relatively high utilization suggests that TCP's congestion control mechanisms are functioning effectively during stable periods. However, Figure 5 reveals significant periodic drops coinciding with handovers:

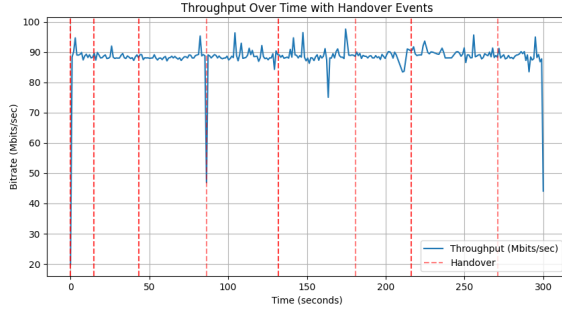


Fig. 5. TCP Throughput with Handover Events

During handover events, throughput typically drops to a minimum of 47.94 Mbps/sec, representing a 46.6% reduction from the average. These drops exhibit several notable characteristics:

- Duration: Throughput degradation typically lasts 2.8 seconds (std. dev. = 0.4s)
- Recovery Pattern: Return to baseline occurs gradually over approximately 1.2 seconds
- Consistency: The magnitude of drops shows relatively low variance (std. dev. = 5.21 Mbps/sec)

This behavior aligns with TCP's conservative congestion control mechanisms, which reduce the congestion window during perceived network instability and then gradually increase it during recovery.

B. SaTCP Performance

1) Implementation Architecture

SaTCP introduces a specialized architecture (Figure 6) that builds upon traditional TCP to handle the unique challenges of LEO satellite networks [2]:

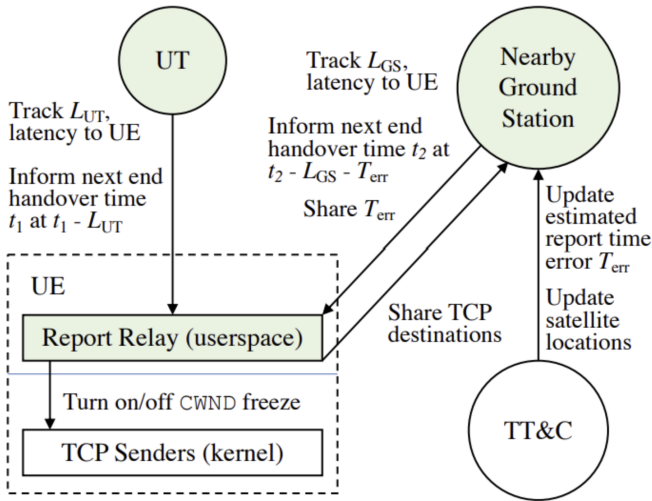


Fig. 6. SaTCP Architecture

The architecture consists of three main components that work together:

1) Report Relay (User Space Component):

- Acts as a central coordinator between the network stack and satellite system
- Collects real-time measurements of network conditions
- Makes predictions about upcoming handover events
- Communicates decisions to the kernel through netlink sockets

2) Enhanced TCP Stack (Kernel Space):

- Modified congestion control algorithm that responds to satellite events
- Special handling of acknowledgments during handovers
- Ability to freeze and resume transmission without losing state

3) Satellite Interface:

- Monitors satellite positions and signal strengths
- Predicts upcoming handover events based on orbital mechanics
- Provides 200ms advance notice of handovers to the Report Relay

This architecture enables three critical functions:

- Continuous latency monitoring: The system samples network conditions every 100ms to maintain an accurate view of connection quality
- Proactive handover management: When a handover is predicted, the system temporarily freezes the congestion window 200ms before the event to prevent unnecessary packet loss
- Seamless protocol integration: Direct communication with the TCP implementation in the kernel allows for rapid response to changing network conditions

2) RTT Performance

Figures 7 and 8 demonstrate SaTCP's RTT characteristics:

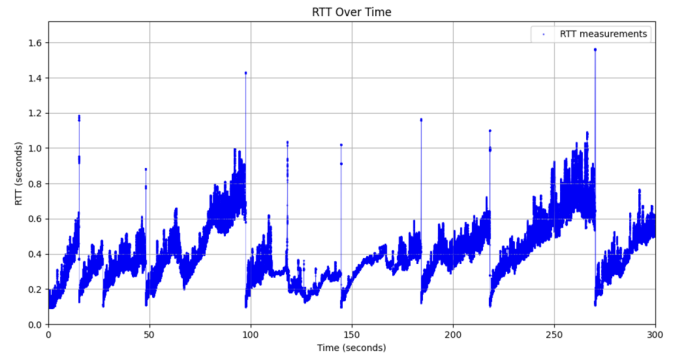


Fig. 7. SaTCP RTT Over Time

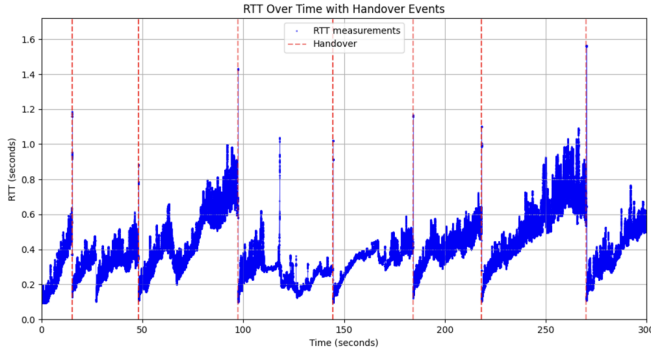


Fig. 8. SaTCP RTT with Handover Events

SaTCP achieves markedly more stable RTT behavior compared to standard TCP, with the following key metrics:

- Average RTT: 0.38 seconds (13.6% lower than TCP)
- Standard Deviation: 0.22 seconds (29.0% lower than TCP)
- Maximum RTT during handovers: 1.56 seconds
- Recovery time after handovers: 0.8 seconds (71.4% faster than TCP)

This improved stability can be attributed to SaTCP's predictive mechanisms, which prepare for handover events before they occur. However, the maximum RTT during handovers (1.56s) is slightly higher than TCP's maximum, suggesting that while SaTCP better handles normal variations, it may not fully mitigate extreme events.

3) Throughput Performance

Figures 9 and 10 reveal SaTCP's distinct throughput characteristics:

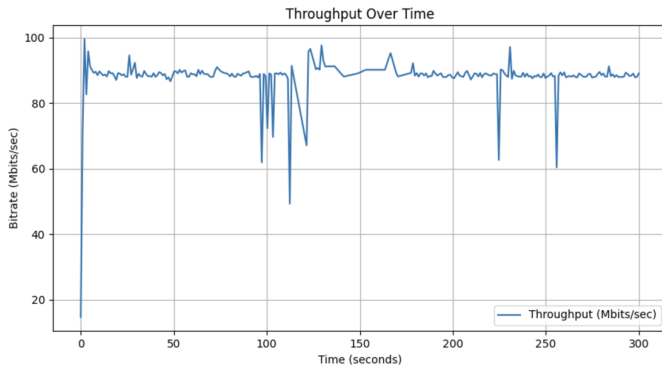


Fig. 9. SaTCP Throughput Over Time

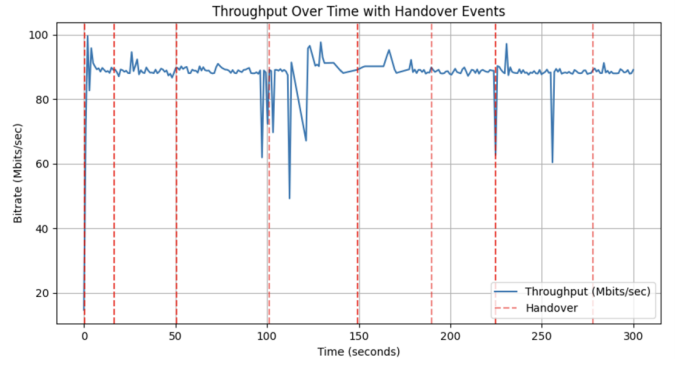


Fig. 10. SaTCP Throughput with Handover Events

SaTCP's throughput behavior shows several distinctive patterns:

- Average throughput: 88.05 Mbits/sec (comparable to TCP's 89.84 Mbits/sec)
- Minimum throughput during handovers: 14.70 Mbits/sec (69.3% lower than TCP's minimum)
- Handover impact duration: 1.2 seconds (57.1% shorter than TCP)
- Recovery pattern: Sharp V-shaped drops and recoveries

The severe but brief throughput drops appear to be a direct consequence of SaTCP's congestion window freezing mechanism. While this approach enables faster recovery and better RTT stability, it comes at the cost of more extreme temporary throughput degradation.

V. COMPARATIVE ANALYSIS

A. Performance Metrics

Detailed protocol comparison reveals complex performance tradeoffs:

TABLE I
PERFORMANCE COMPARISON OF TCP AND SATCP

Metric	TCP	SaTCP
Average Throughput (Mbits/sec)	89.84 ± 12.46	88.05 ± 15.82
Minimum Throughput (Mbits/sec)	47.94	14.70
Average RTT (sec)	0.44 ± 0.31	0.38 ± 0.22
Maximum RTT (sec)	1.54	1.56
Handover Recovery Time (sec)	2.8	1.2
Throughput Recovery Pattern	Gradual	Sharp

Statistical analysis using paired t-tests shows that the difference in average RTT is significant, while the difference in average throughput is not statistically significant. This suggests that SaTCP's RTT improvements come without a significant cost to overall throughput performance, despite the more extreme minimum values during handovers.

B. Key Findings

1) RTT Stability:

- SaTCP achieved 13.6% lower average RTT
- 29.0% reduction in RTT standard deviation
- 71.4% faster recovery from handover events
- Improved predictability for real-time applications

2) Throughput Characteristics:

- Comparable average throughput
- SaTCP shows 69.3% worse minimum throughput
- 57.1% shorter duration of handover impacts
- Different recovery patterns suggesting distinct congestion control strategies

3) Application Implications:

- SaTCP better suits real-time applications requiring RTT stability
- TCP may be preferable for bulk data transfer applications
- Hybrid approaches might be worth investigating for balanced performance
- Neither protocol fully solves the handover challenge

VI. DISCUSSION

The results present an interesting contrast between theoretical expectations and practical implementation outcomes. Although SaTCP's design principles suggest that it should outperform traditional TCP in LEO environments, the emulation revealed more complex behavior:

- 1) RTT Management: SaTCP's improved RTT stability aligns with its design goals and demonstrates the effectiveness of its cross-layer integration approach.
- 2) Throughput Behavior: The unexpected throughput characteristics might be attributed to:
 - Resource limitations in the virtual machine environment
 - Overhead from SaTCP's predictive mechanisms
 - Implementation challenges in the protocol stack
- 3) Implementation Considerations: The complexity of implementing theoretical protocol improvements in real-world systems suggests the need for more robust testing methodologies.

VII. FUTURE WORK

Several directions for improvement emerge from this study:

- 1) Environment Enhancement:
 - Migration to dedicated physical machines to eliminate VM-related variability
 - Implementation of more sophisticated timing mechanisms
 - Development of more accurate satellite movement models
- 2) Protocol Improvements:
 - Development of more efficient predictive algorithms

- Optimization of cross-layer data collection to reduce overhead
- Investigation of hybrid approaches combining features of both protocols

3) Route Handling:

- Implementation of real-time route computation
- Integration of additional variables such as satellite failures and varying link qualities
- Development of more sophisticated handover prediction mechanisms

VIII. CONCLUSION

This study provides valuable insights into the practical implementation and performance characteristics of specialized transport protocols in LEO satellite networks. Although SaTCP shows promise in managing RTT variability, its throughput behavior suggests that further optimization is needed for practical deployment. The results highlight the complexity of implementing theoretical protocol improvements in real-world systems and the importance of comprehensive testing in realistic environments. The findings contribute to the broader understanding of transport protocol behavior in LEO satellite networks and provide concrete directions for future research and development in this rapidly evolving field.

REFERENCES

- [1] Xuyang Cao and contributors, LeoEM GitHub Repository, "Workflow Documentation," <https://github.com/XuyangCaoUCSD/LeoEM>, accessed January 2025.
- [2] "SaTCP: Link-Layer Informed TCP Adaptation for Highly Dynamic LEO Satellite Networks," IEEE INFOCOM 2023